A-A-55007
24 September 1991

COMMERCIAL ITEM DESCRIPTION

"V" BLOCKS AND CLAMPS, COMBINATION TOP AND END "V" GROOVES

The General Services Administration has authorized the use of this commercial item description in preference to the Type IV, Style A of A-A-51150.

Abstract. This description covers "V" blocks and clamps with a "V" groove on the top and one end. The blocks are used in the inspection of parts and other tool room operations. The "V" blocks and clamps shall be similar to that shown in figure 1.

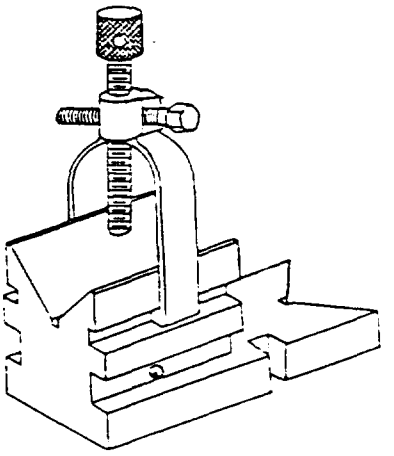


FIGURE 1. "V" Block and clamp combination top and end "V" groove.

Beneficial comments recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5213.

AMSC N/A FSC 3460

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Salient characteristics.

1. Design. The "V" blocks shall be provided with slots for single level clamping. There shall be no interference from the clamping device when turned on the side . Each "V" block shall be provided with a 90 degree "V" groove, 45 degrees from center cut and extending the entire length of the "V" block. At the apex of the groove, a slot shall extend the length of the block providing for chip relief. One end of the block shall have a projection or step and shall also contain a "V" groove, relief slot, and clamping slots. Unless otherwise specified (see ordering data), the "V" blocks shall be furnished in pairs with V's matched. The "V" blocks shall be plainly marked to identify them as a matched pair.

2. Size. The "V" blocks furnished shall be 3-1/2 long by 1-7/8 inch wide by 1-7/8 inches high, with a diametrical capacity of 1 5/16 inches. All dimensions shall be +0.03/-0.00 inch.

3. Materials. The blocks shall be manufactured of alloy steel.

4. Hardness. The hardness for the "V" blocks shall be not less than 60 and not more 65 on the Rockwell "C" scale when tested in accordance with ASTM E 18.

5. Tolerance. The "V" groove shall be central with the sides and square and parallel with all surfaces within 0.0003 inches. All surfaces shall be square with each other within 0.0002 inches.

6. Surface finish. The surface finish of the "V" blocks shall have a roughness average (RA) of not greater than 32 microinches when measured in accordance with ASME B46.1. Edges of all exterior surfaces shall be free from sharp edges, fins and burrs.

7. Clamp. The clamping device shall be of the bale type similar to that shown in figure 1. The clamping device shall be designed for use for the top and end clamping slots. The screws for the clamping device shall be right hand threads conforming to FED-STD-H28.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. During the manufacture of the "V" blocks and clamps, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract, shall not be used or emitted.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract order.

NOTES

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Size and quantity required.

c. If other than matched pairs (see para 1).

Federal Standard FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.

ASTM E 18 is available from American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

ASME B46.1 is available from The American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.

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